

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Naveen

May 2024

Course Code: MAT102R01

Course: ENGINEERING MATHEMATICS - II

QP No. :U175-2

Duration: 3 hours

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. Find the curvature of the curve $y = e^x$ at $(0, 1)$.
2. Determine the envelope of the family of straight lines $y = mx + \frac{a}{m}$ where m is the parameter.
3. The profit equation of a company is given by the function $f = x^3 - y^2 - 3x$ and hence determine its maximum profit value.
4. Check whether the differential equation $e^y dx + (xe^y + 2y) dy = 0$ is exact and hence solve it.
5. Find the primitive of $p = \log(px-y)$.
6. Find the deflection $y(x)$ of bending of a beam fixed at it two ends, satisfying $\frac{d^4 y}{dx^4} - y = 0$.
7. If ϕ is a surface then what can you infer from $\nabla \phi$?

8. If $\vec{r}$ is a position vector then find its divergence and curl values.
9. State Stoke's theorem.
10. Use Stoke's theorem to evaluate $\int_C e^x dx + 2y dy - dz$ where C is the curve $x^2 + y^2 = 4$ and $z = 2$.

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. a) Determine the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (7)
- b) Identify the envelope of family of straight lines which slides with its ends on two perpendicular axes such that sum of the n th powers of the intercepts on the axes is constant. (8)

(OR)

12. a) Determine the evolute of parabola $y^2 = 4ax$, treating it as envelope of normals. (7)
- b) The pressure P at any point of the space (x, y, z) is given by $P = 400xyz^2$, determine the highest pressure at the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (8)
13. a) Solve $p^2x + py - y^4 = 0$ where $p = \frac{dy}{dx}$. (7)
- b) A body of temperature 80°F is placed in a room of constant temperature 50°F at time $t = 0$. At the end of 5 minutes the body has cooled to a temperature of 70°F . (i) Find the temperature of a body at the end of 10 minutes. (ii) After how many minutes will the temperature of the body be within 1°F of the constant 50°F temperature of the room. (8)

(OR)

14. Solve the following system:

$$\frac{dx}{dt} + 4x + 3y = t$$

$$\frac{dy}{dt} + 2x + 5y = e^t$$

15. a) Find the angle between the surfaces

$$x^2 + yz = 2 \text{ and } x + 2y - z = 2 \text{ at } (1, 1, 1).$$

(7)

b) Show that

$$\vec{F} = (y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k} \text{ is both solenoidal and irrotational.}$$

(8)

(OR)

16. a) Identify the values of $\text{div}(\vec{r}''\vec{r})$ and $\text{curl}(\vec{r}''\vec{r})$ where $\vec{r}$ is a position vector.

(7)

b) Prove that $\text{div}(\text{curl } \vec{f}) = 0$ and $\text{curl}(\text{grad } \phi) = \vec{0}$.

(8)

17. Verify Green's theorem in the xy plane for

$$\int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy \text{ where } C \text{ is a closed curve bounded by}$$

$$y = x^2 \text{ and } y^2 = x.$$

(OR)

18. Verify Gauss divergence theorem for $\vec{f} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ over the surface S of the cuboid formed by the planes $x = 0, x = a; y = 0, y = b; z = 0, z = c$.

Answer the following

1 x 20 = 20 Marks

19. a) From any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ perpendiculars are drawn to the axes and foot of these perpendiculars are joined. Identify the envelope of family of lines thus formed. (10)

b) A condenser of capacitance C is charged through a resistance R by steady voltage E . Show that the charge q on the condenser at any time t is $EC(1 - e^{-\frac{1}{RC}t})$ if $q = 0$ at $t = 0$ and hence show that the current $i = \frac{E}{R} e^{-\frac{1}{RC}t}$. (10)
